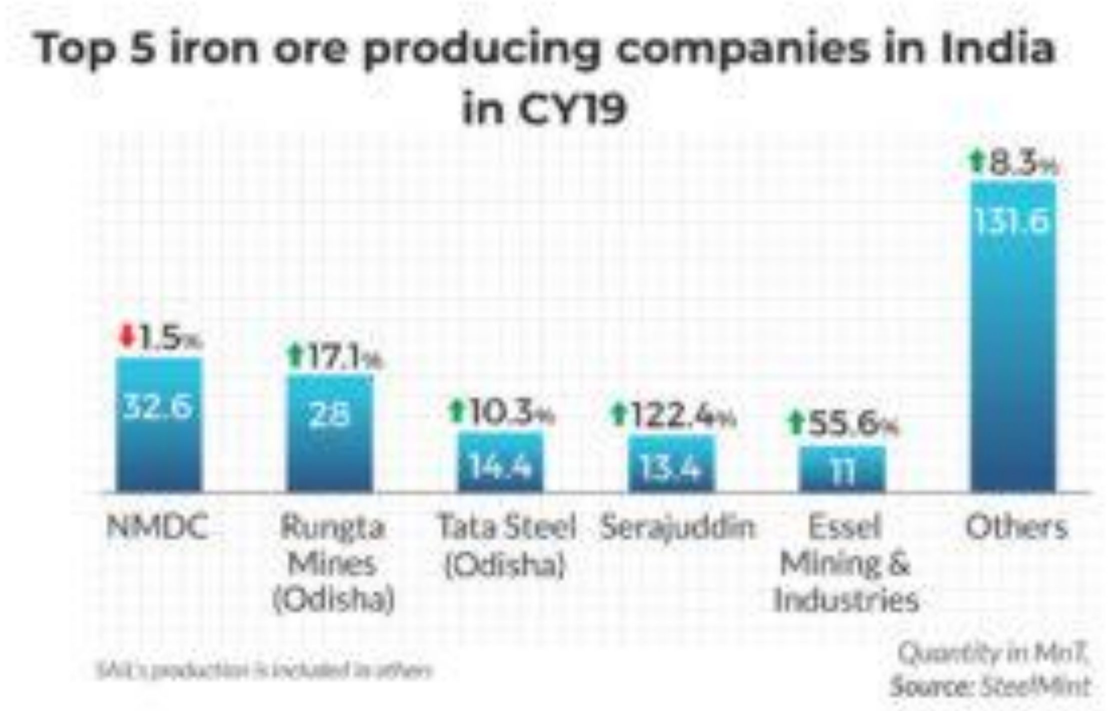
Cost of mining



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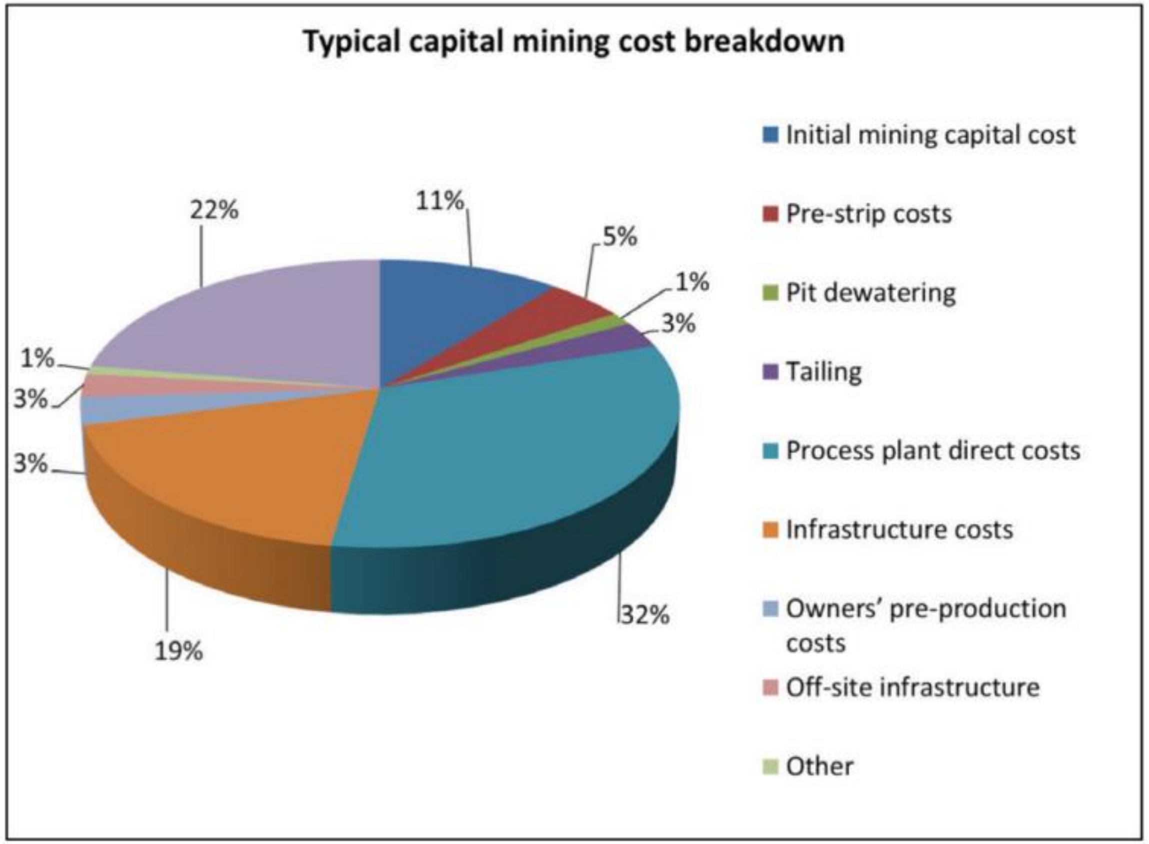

Typical capital mining cost breakdown
22%
11%
5%
1%
3%
1%
3%
3%
19%
32%
Initial mining capital cost
Pre-strip costs
Pit dewatering
Tailing
Process plant direct costs
Infrastructure costs
Owners' pre-production costs
Off-site infrastructure
Other

Contents

- Capital and operating costs;
- Factors affecting operating cost;
- Methods of estimating future costs;
- Standard cost and forecast

There are a number of different types of costs which are incurred in a mining operation. There are also many ways in which they can be reported. Three cost categories might be:

- Capital cost;
- Operating cost;
- General and administrative cost (G&A).

The capital cost in this case might refer to the investment required for the mine and mill plant. The operating costs would reflect drilling, blasting, etc. costs incurred on a per ton basis. The general and administrative cost might be a yearly charge.

The G&A cost could include one or more of the following:
- Area supervision; - Mine supervision; - Employee benefits; - Overtime premium;
- Mine office expense; - Head office expense; - Mine surveying; - Pumping;
- Development drilling; - Payroll taxes; - State and local taxes; - Insurance;
- Assaying; - Mine plant depreciation.

The capital and G&A costs could be translated into a cost per ton basis just as the operating costs. The cost categories might then become:
- Ownership cost;
- Production cost;
- General and administrative costs.

The operating cost can be reported by the different unit operations:
- Drilling; - Blasting; - Loading; - Hauling; - Other.

The 'other category' could be broken down to include dozing, grading, road maintenance, dump maintenance, pumping, etc. Some mines include maintenance costs together with the operating costs. Others might include it under G&A. Material cost can be further broken down into components. For blasting this might mean:
- Explosive; - Caps; - Primers; - Downlines.

The operating cost could just as easily be broken down for example into the categories:
- Labor;
- Materials, expenses and power (MEP);
- Other.

At a given operation, the labor expense may include only the direct labor (driller, and driller helper, for example). At another the indirect labor (supervision, repair, etc.) could be included as well.
There are certain costs which are regarded as 'fixed', or independent of the production level. Other costs are 'variable', depending directly on production level. Still other costs are somewhere in between.
Costs can be charged against the ore, against the waste, or against both.
For equipment the ownership cost is often broken down into depreciation and an average annual investment cost. The average annual investment cost may include for example taxes, insurance and interest (the cost of money).

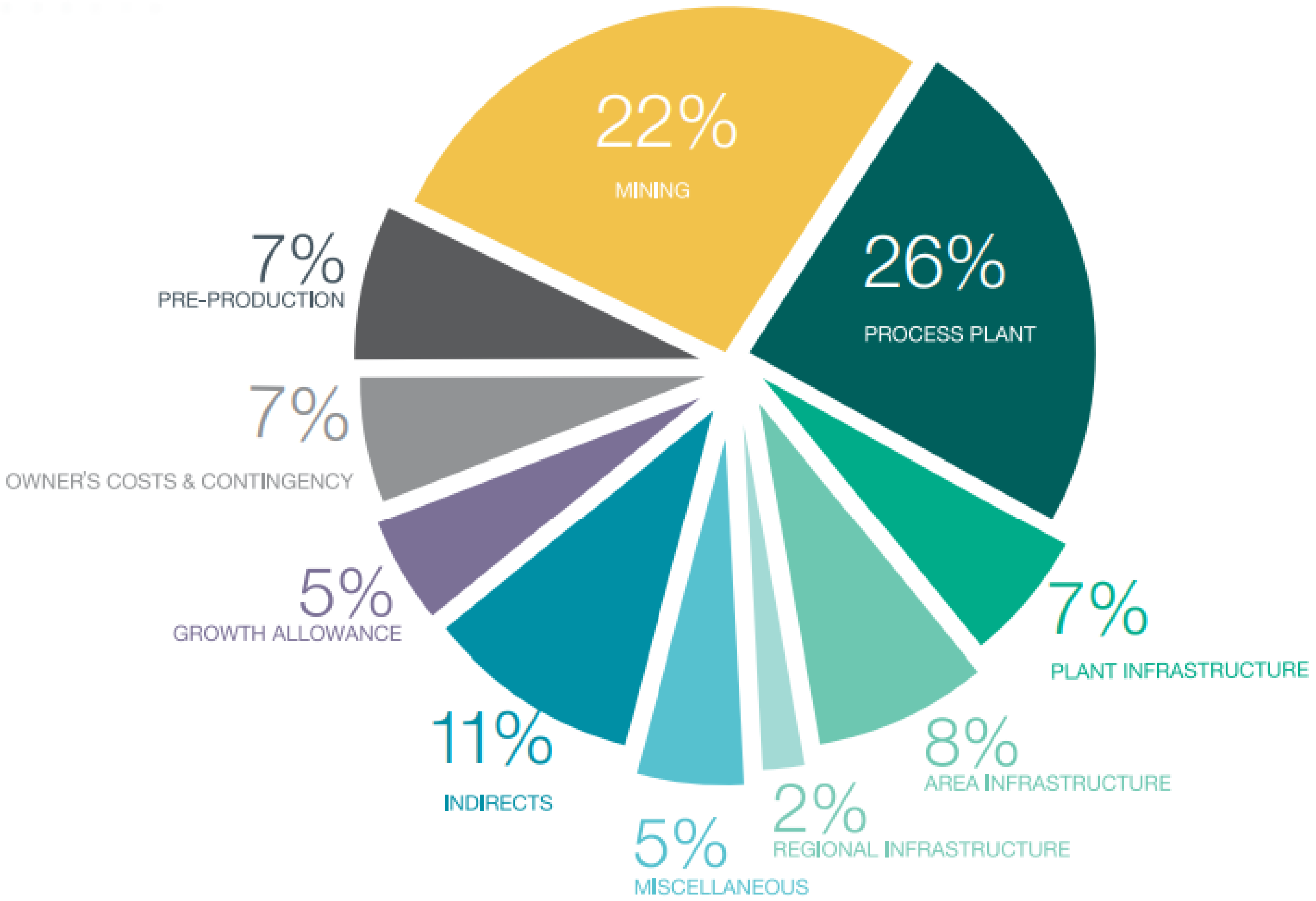
Capital cost mining equipment

Equipment	Size	Price ($1000)
1. Rotary drills, electric	Hole size (ins)	Price ($1000)
	9 to 12-1/4	1,842
	9-7/8 to 16	2,898
	10-3/4 to 17-1/2	2,983
2. Graders	Blade width (ft)	Price ($1000)
	12	359
	14	465
	16	765
	24	1775
3. Wheel loaders	Bucket capacity (yd^3)	Price ($1000)
	9	793
	11	1,290
	16	1,752
	18	1,774
	21	2,224
	26	4,633
	33	5,924
	53	7,652
4. Electric cable shovels (rock)	Bucket capacity (yd^3)	Price ($1000)
	11	3,397
	16	4,458
	20	5,307
	27	8,793
	34	9,975
	55	12,655
5. Trucks, rear dump (mechanical drive)	Capacity (tons)	Price ($1000)
	40	630
	60	881
	65	895
	70	1,011
	100	1,288
	150	2,120
	170	2,517
	200	3,105
	250	3,912
	360	4,587
	380	6,050
6. Trucks, rear dump (electric drive)	Capacity (tons)	Price ($1000)
	173	2,521
	200	2,835
	280	5,583
	314	3,935
	360	6,218
7. Crawler tractors (dozers) (with ripper)	Horsepower	Price ($1000)
	260	703
	310	714
	330	735
	350	852
	410	1,047
	570	1,137
	850	2,086

Capital and operating costs

- Mining capital cost (MCC) is one of the essential criteria for assessing the feasibility of an open-pit mine (or underground mine). The MCC heavily influences the net present value (NPV) of the projects over the lifetime of the mine.
- Cost of capital represents a hurdle rate that a company must overcome before it can generate value, and it is used extensively in the capital budgeting process to determine whether a company should proceed with a project.
- The estimate includes all the necessary costs associated with engineering, drafting, procurement, construction, construction management, commissioning of the processing facility and associated infrastructure, mining infrastructure, first fills of plant reagents, consumables and spare parts.
- The estimate is based upon preliminary engineering, material take-offs and budget price quotations for major equipment and bulk commodities.
- Unit rates for installation were based on market enquiries specific to the MRP and benchmarked to those achieved recently on similar projects undertaken in the Australian minerals processing industry.
- The overall capital estimate is considered to be with an estimate accuracy of ±10 to 15%. The basis of estimate is summarised in Table.

Capital Cost Breakdown – by Work Breakdown Structure



The open pit mine capital costs include the following:

- Heavy mobile equipment (HME)
- Minor equipment
- Explosives and diesel
- Mining software and systems
- Spare shovel bucket
- Spare FEL bucket
- Truck tray re-liner package
- Dewatering
- Pit dispatch system
- Global positioning system for mine surveying
- Blasting contractor set-up
- Maintenance system
- Pit power distribution
- Training simulator

General and Administration Operating Cost

- Executive Management
- Human Resources and Training
- Business Readiness and Integration
- Health, Safety and Security
- Regional Development & Communications
- Government Relations
- Commercial

Underground Capital Cost

Underground capital costs are divided into the following two categories:
Capital Development – includes all preproduction and production build-up development and infrastructure required to initiate production and undercut the cave

Sustaining Capital – includes work required for long-term development, facilities, or systems to sustain full production operations, but not direct production cost

The components of capital development and sustaining capital are further defined below.

- Drifting and Excavating
  - Rock handling – truck haulage, conveyor drift, and crosscut development and associated rock handling excavation.
  - Ramps and accesses – access ramp, level access, and level development.
  - Fixed facilities drifting – infrastructure excavation (i.e. material staging area, shops, lunchrooms, etc).
  - Extraction and undercut drifts – perimeter drift, panel drift, drawpoint drift, and drawbell drift development.
  - Ventilation drifts and accesses – dedicated intake and exhaust airways (lateral only).
  - Mass excavations – large excavations required for facilities and installations.
- Shafts and Raises
  - Intake and exhaust raise development.
  - Ore pass and waste pass development.
  - Service and production shaft development.
  - Ventilation shaft development.
- Equipment and Infrastructure
  - Truck haulage infrastructure – supply and installation of truck haulage-related fixed equipment and infrastructure.
  - Ore flow infrastructure – supply and installation of conveyors grizzlies, ore chutes, crushers, etc.
  - Drawpoint construction – drawpoint related excavation and construction activities, including construction of drawpoints, slot raise excavation, drawbell and undercut drilling and blasting, and undercut swell mucking.
  - Mobile equipment – production LHDs, production drills, haulage trucks, development jumbos, bolters, trucks, utility vehicles, etc. This includes the initial supply, materials for major rebuilds, and replacement of mobile units.
  - Fixed facilities infrastructure – ventilation facilities, pumping systems, shops, offices, lunchrooms, storage areas, fuel and lube areas, explosives magazine.

Operating Cost

- Mining
  - Open Pit
  - Underground
- Processing
- Tailings
- G&A
- Other
  - Government Fees and Charges
  - Management Fees

The main operating cost driver is haulage, followed by loading and loading

Underground Operating Costs

Operating costs include direct and indirect costs associated with the production of ore. Production is defined as pulling ore from drawpoints and delivering it to the main overland conveyor and all activity directly supporting this process. Operating costs include the following items:

- Production
  - Production mucking – production haulage by LHDs from drawpoint to extraction level ore passes.
  - Truck haulage and ore flow – truck operation, chute maintenance, apron feeder maintenance, crushing and conveying system operation, and conveyor spill cleanup.
  - Secondary breaking – fixed and semi-mobile rock breakers, rock breaker operators, breaking of drawpoint oversize, and relieving low and medium hang-ups.
- Ground Repair
  - Drawpoint repair.
  - Sill concrete repair.
  - Ore pass repair.
- Indirects – costs not directly allocated to production.

Factors affecting operating cost

Staff

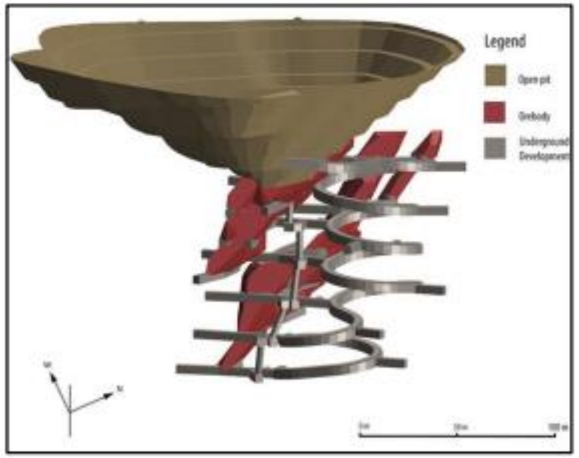
Camp and Travel

Labour

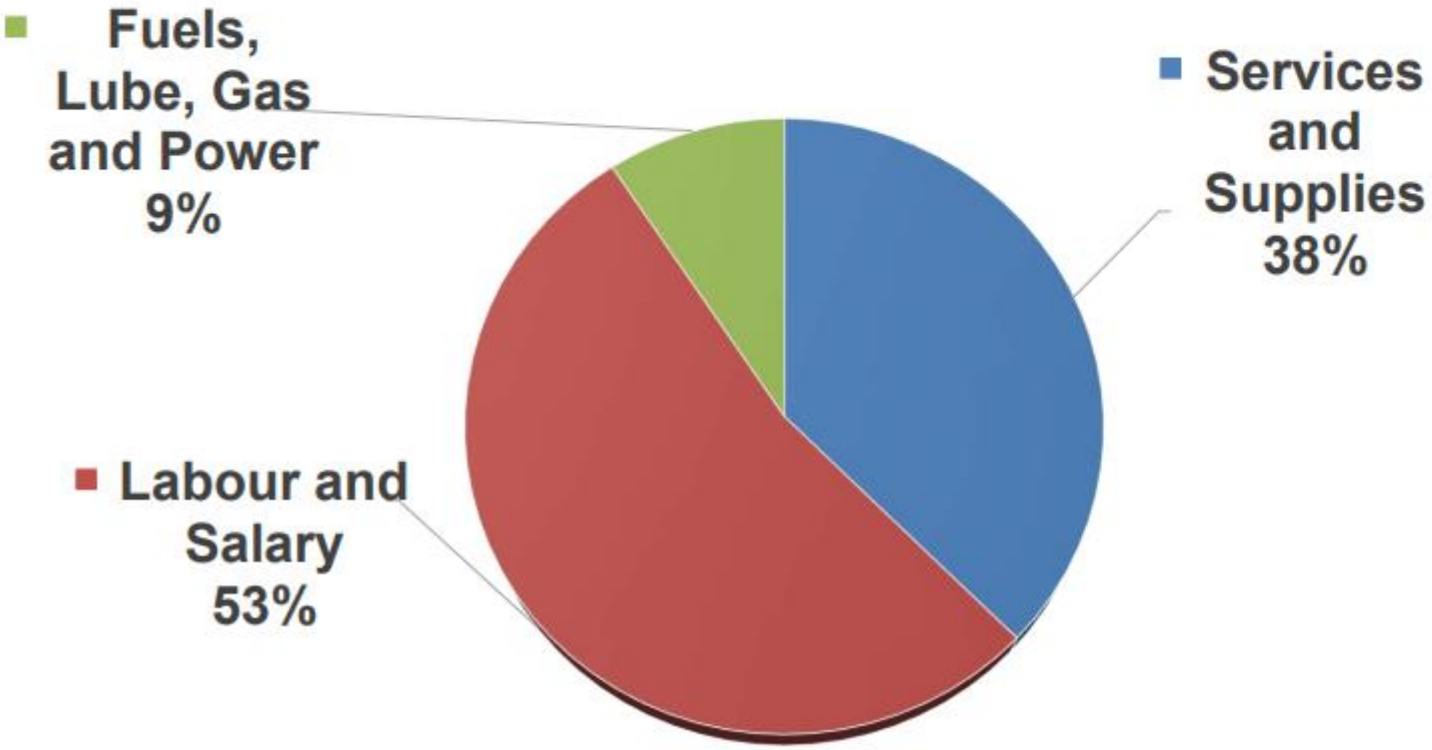
Mobile Equipment Parts

Services

Processing



Supplies



Traditional Costing – Unit Based

Traditional cost accounting methods simply allocate costs, down onto the cost objects without considering any 'cause and effect'.

Breakdown by Labour costs, Materials and Supplies

Activity Based Costing

- ABC is an accounting method that allows mines to gather data about their operating costs

- Costs are assigned to specific activities including:
  - Drilling
  - Engineering
  - Truck Haulage

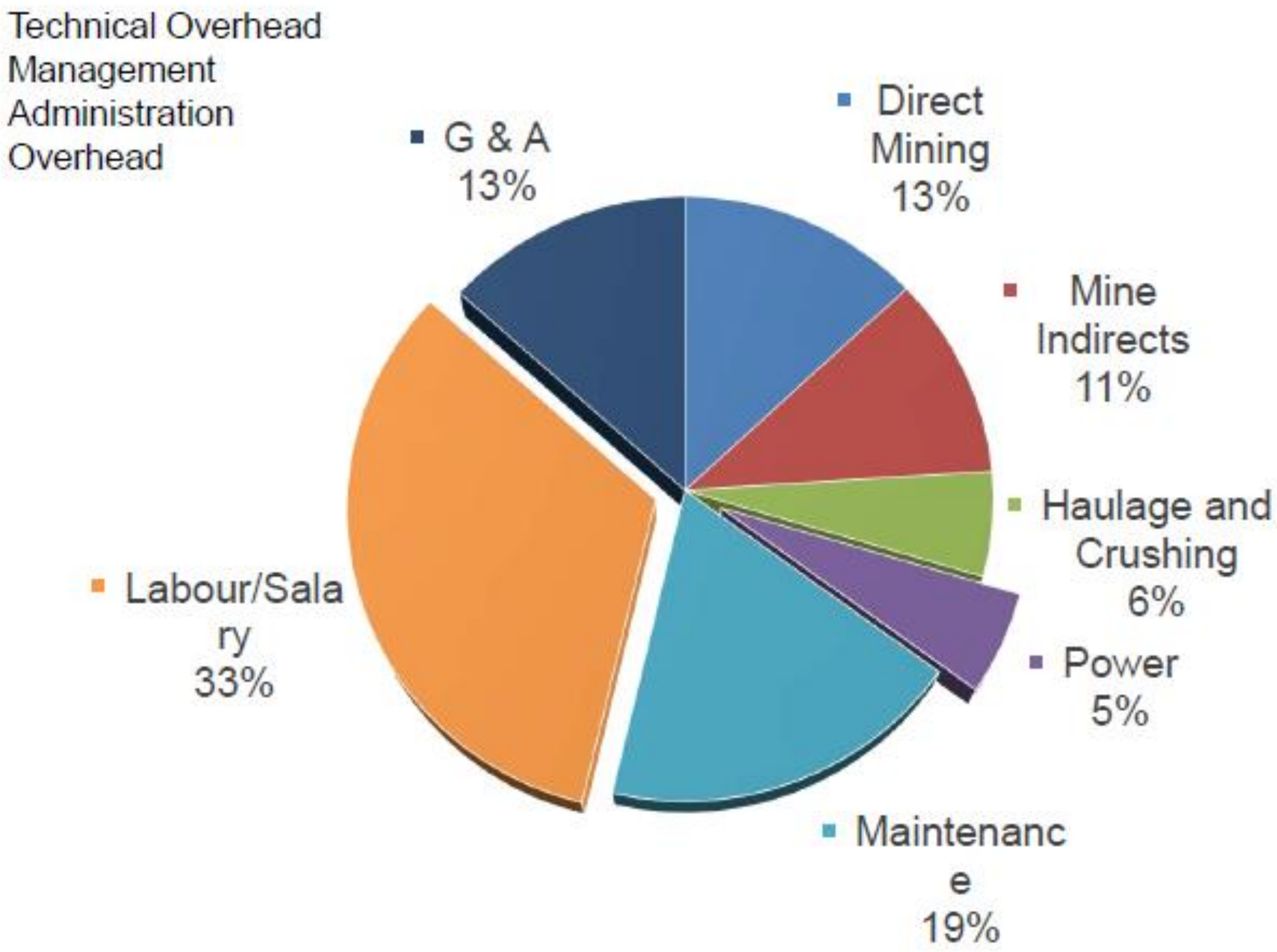
- activities are associated with the generated overall average tonnes produced per day.

$$\text{Activity-Based per (Unit) tonne} = \frac{\text{Total Activity (Process) Cost)}}{\text{Total Number of units (tonnes)}}$$

- Enables the mine to decide which activity may be increasing their profitability, and which are contributing to losses.

- able to generate data to create an improved budget and gain a greater overall understanding of the expenses that are required to keep the mine running.

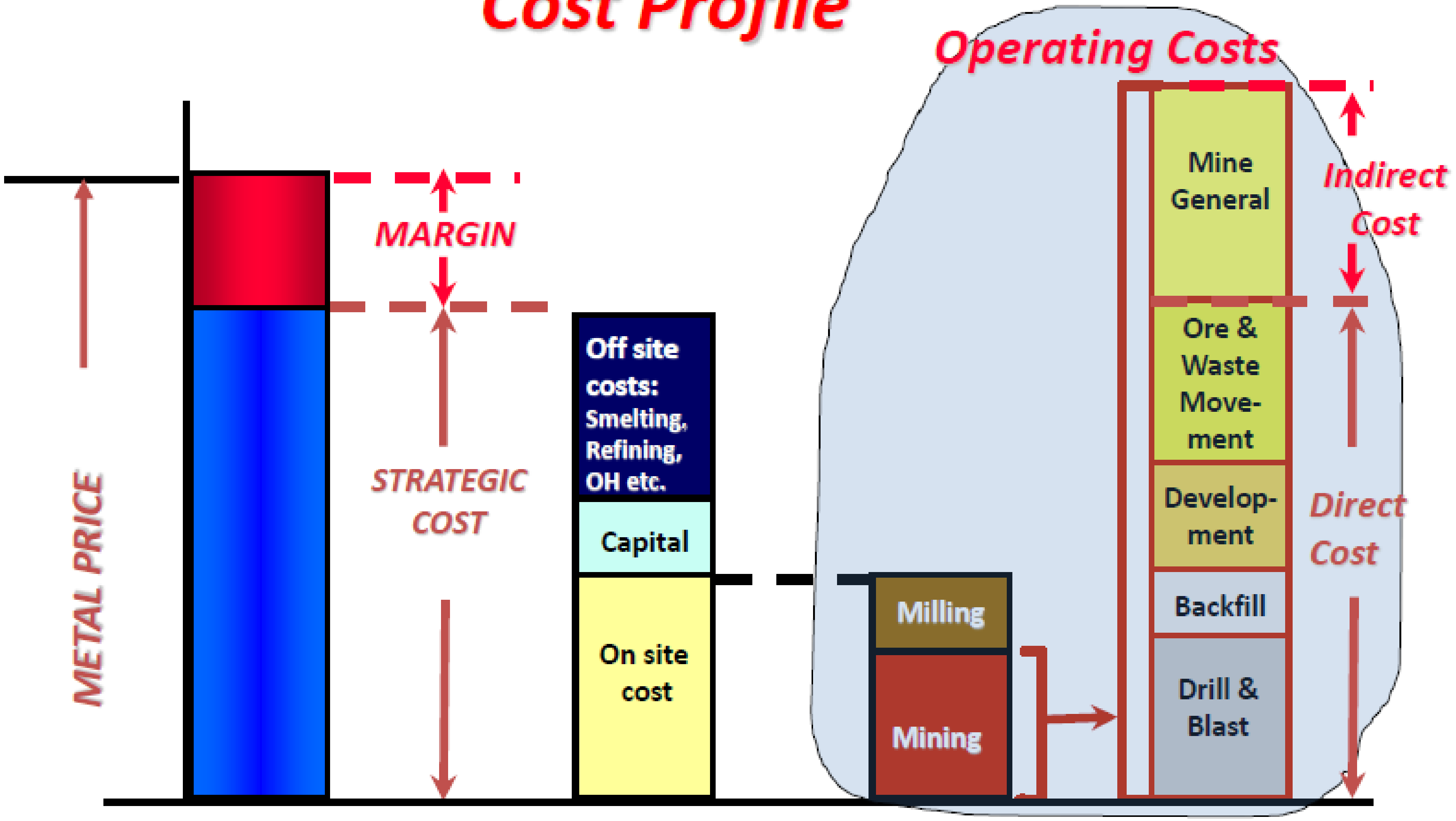
Mine Cost in details



Lateral Development
Shotcreting
Cable Bolting
Primary Ground Support
Secondary Ground Support
Hydraulic Fill
ITH Drilling
LHD Mucking
Production Blasting
Standard Raising
Top Hammer
UG Truck Haulage
UG Rail Haulage

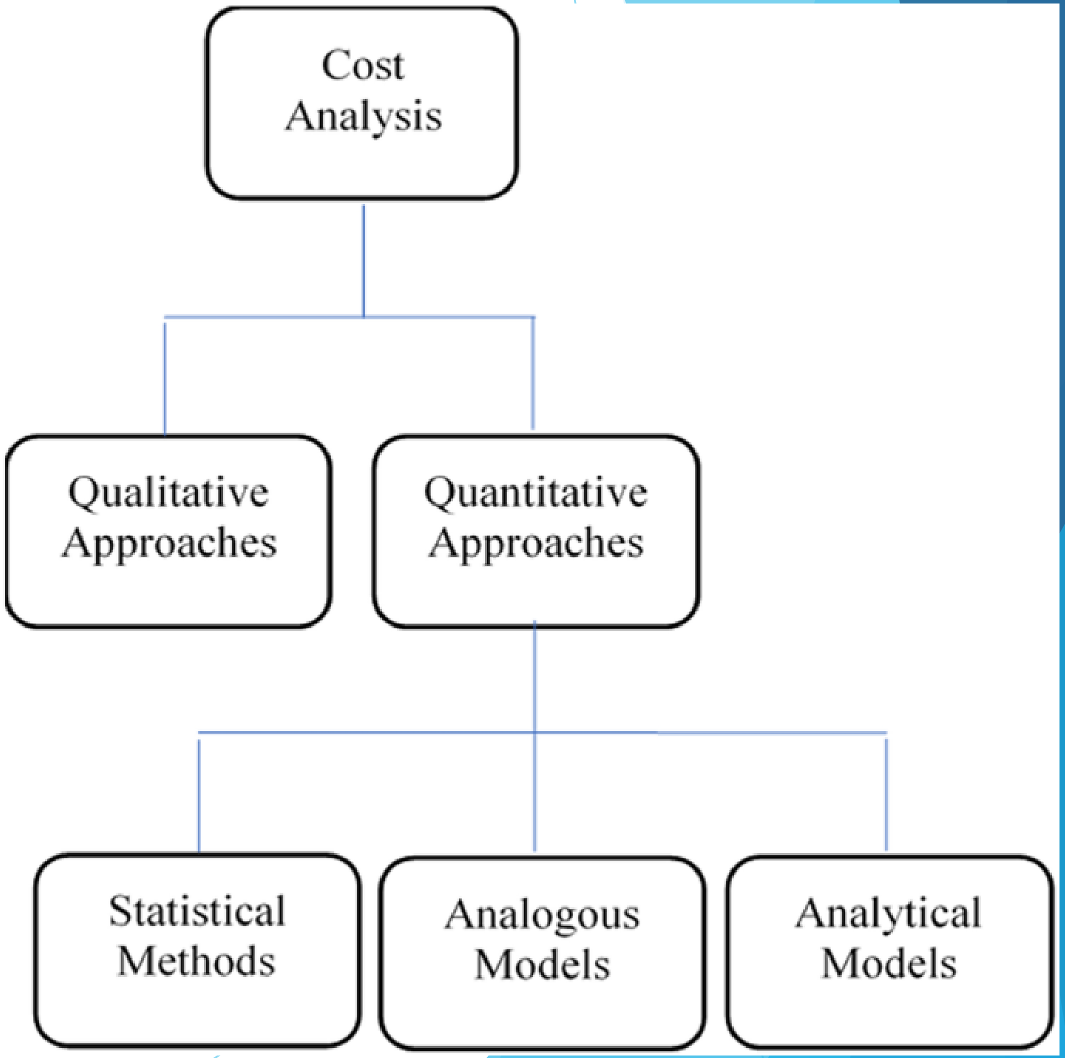
Mining Indirects
Raise boring
Muck Circuit
Hoist and Shaft Services
Training
Roadway Upkeep
Supplies Handling
Services - Utilities
Plant Security

Cost Profile



Methods of estimating future costs

Qualitative approaches are based on estimator’s knowledge of the project, the scope of work, and influencing factors and are divided into two classes: expert judgment and heuristic rules. Expert judgment depends on the good or bad results of the past estimations based on judgment



Quantitative approaches can be defined as methods relying on the process of collecting and analysing historical data and applying quantitative models, techniques, and tools to estimate the project’s cost. Quantitative cost estimating approaches are classified into three main categories: statistical, analogous, analytical ones.

Statistical methods in cost estimation

- Statistical methods, are based on formulas or other alternative approaches to establish a causal correlation between final costs and its corresponding characteristics.
- Parametric cost estimating methods evaluate the cost through regarding characterizing parameters like mass, volume, and cost without considering little details.
- There are three types of parametric cost estimation methods as follows:
- The method of scales / assuming that the cost and considered parameters are interrelated through a linear function. This method is applicable in prevailing technologies, which simple products of different sizes are produced. Evaluating the most influencing technical parameters is the prerequisite of this method. Thereafter, this evaluation is compared with those of finished projects, which makes this method a combination of analogous and parametric approaches.
- *Statistical models* In this method, the activities are divided into major different scopes through, which the final mathematical formulae is constructed. This model is composed of three main data types: Technical specifications, Relationships between the data and final variables, Constants

- *Cost estimation formulae (CEF)* CEF is a mathematical relationship between the final cost and a limited set of technical parameters. The major parameter categories are as follows:
- • *Physical values* According to functional description
- • *Dimensioning values* According to solution description

The most probably prevailing parametric methods are regression analysis and optimization techniques.

- Parametric cost estimation methods are faced with different drawbacks, which some of them are described as follows; through application of these methods, different results are the sole issue without giving a vision about the origin of them.
- On the other hand, lack of necessary parameters during early stages will result in uncertainty of the results. In addition, the designer should be aware of the influence of each parameter on the final cost. CEFs in particular, are incapable of solving specific cases. Sometimes also, there is a need to obtain results of regression analysis in four or five similar cases to reach to the most reliable cost. Despite all these disadvantages, they are considered as useful cost estimation methods due to their rapidity of execution

various methods (ANN, Fuzzy NN, SVM, PSO, RBF, RA, CBR, PSO, Decision Tree, AHP, Monte Carlo, fuzzy logic)

Analogous models

- Analogous models are based on similar past cases, which are reused and adjusted in different cases.
- This similarity is due to functional or geometrical homogeny between cost structures, which are alike.
- Analogous methods are known to be the simplest method of estimating through. The cost of projects is estimated in compared to their similar completed projects that are available as a historical database.
- Thus, project managers have to consider the most available parameters to include in their process of estimating to reach better results; however, this method is a kind of rough estimate, which is easy to use, but with lower levels of complexity and accuracy as well.

Analytical models

Analytical models instead, are the process of estimating costs by accurately defining the cost corresponding to each processing phase attribute in details, and afterwards using a bottom-up approach for aggregating the project total cost, thus this approach is leading to a more accurate result

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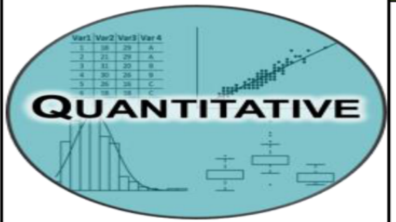</td><td colspan="5">Parametric</td></tr>
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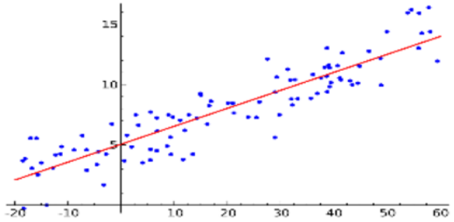
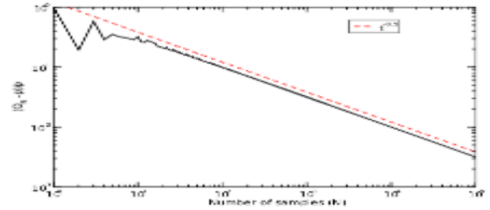
<tr><td colspan="2"></td><td></td><td colspan="2"></td></tr>
<tr><td colspan="5">Analogous</td></tr>
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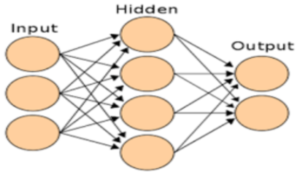
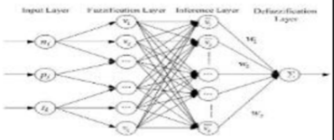
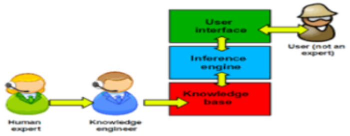
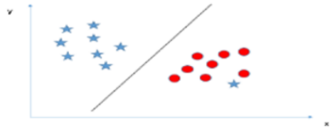
<tr><td></td><td></td><td></td><td></td><td></td></tr>
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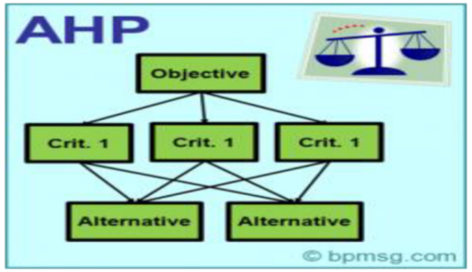
<tr><td></td><td colspan="5"></td></tr>
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Standard cost and forecast

- A standard cost is described as a predetermined cost, an estimated future cost, an expected cost, a budgeted unit cost, a forecast cost, or as the "should be" cost.
- When standard costs are used in a manufacturing setting, a product's standard cost for a future accounting period will consist of the following:
- Direct materials: a standard quantity of each material and a standard cost per unit of material
- Direct labor: a standard quantity of labor and a standard cost per hour of labor
- Manufacturing overhead: a budget for the fixed overhead, the standard variable overhead rate, and the standard quantity for applying a fixed and variable overhead rates

Cost: [Cost estimator.pdf]

Three industry focus areas of productivity

	Labour	Investment Capital	Operating Asset
What	Output generated per hour of work undertaken. It is measured in dollars of gross value added (GVA) per hour.	Real output($) per unit of capital services($).	Real output($) per resource.
How	• Identifying sources of talent • Disciplined workforce planning • Develop EVPs with a focus beyond monetary incentives.	• Ensure internal rigour in the CAPEX review process • Get the parameters for financial modelling right • Get the level of investment right	• Maintenance • Reliability • Asset utilisation • Operational cycle times • Operations planning and control
Why	Skills shortages have been driving rising labour costs.	Increase investment in profitable assets.	Increase margin

Low productivity in one stage of the value chain filters through to the subsequent stages

Core Activities

Exploration

Extraction

Processing

Logistics

Support Services

Discreet Activity Solutions

Exploration:
- Program Planning
- Drilling Efficiency
- Remote Logistics

Extraction:
- Pit Optimisation
- Extraction Methodology
- Fleet Productivity & Fit

Processing:
- Plant Optimisation
- Maintenance Strategy & Execution
- Capacity Planning

Logistics:
- Intermodal Optimisation
- Procedural Review
- Capacity Analysis

Support Services:
- Benchmarking
- Process Mapping
- Resource Allocation

Cross Activity Solutions

←

- Outsourcing
- Warehouse & Logistics Optimisation
- Supply Chain Optimisation
- Shared Services Analysis & Implementation

- Process Bottleneck & Constraint Analysis
- Cost Reduction
- Application Implementation

→

Getting one stage wrong leads to inefficiencies in the following stages of the extraction process

Process

Drill	Blast	Load	Haul
Prepare ground for blasting	Break the ground	Load broken ground into trucks	Haul ore and waste

Blast

Issues can add 10 to 25% to downstream costs through:

- Damage to fleet, ore body, mine plan
- Inefficient processing
- Waste of explosive
- Re-work

Load

There needs to be sufficient stock of broken ground to:

- Deliver preferred material blend to processing plant
- Achieve sizing within specification
- Ensure effective cycle time and utilisation of fleet

Inefficient loading can consume large amounts of capital and drive the operation up the cost curve

Haul

Critical success factors:

- Loading efficiency - matching haul truck size with loader size
- Operational efficiency - matching fleet disposition and mine plan
- Avoiding queuing at the loader or dump/crusher
- Avoiding double-dipping from inefficient loading practices
- Reducing tyre impacts from road conditions
- Reliability from schedule and plan compliance

Thanks